

1 Initial Notes

So I read all the feedback. The most common issues were me going too fast, the problems given, my explanations to the problems, and not enough variations of problems. So, to remedy this, I'm basically going to go rogue and ditch a lot of the curriculum that I was suggested to follow. Personally, I think a lot of the problems are ambiguous and their solutions are random. So I'm going to try to focus on one or two types of problems per class, lecture on them with several examples, and then give you worksheets to work on. I will choose material whose solutions are more straight forward.

Concerning the exams, there are several students who have to take the exam still, so you won't get them back until next week. I won't have the homework back until next week either. Instead of grading this past weekend, I focused on restructuring the course so it flows a bit better.

2 Stretches

1. Assign True or False to each sentence
 - (a) Sentence 5 is false.
 - (b) Sentence 1 is false.
 - (c) Sentence 4 is true.
 - (d) Only one of these sentences is false.
 - (e) Sentences 2 and 3 are both true or both false.
2. Merlin the Magician encountered three individuals on the road. Though he didn't know them, he knew one was a knight. He also knew that spies always lied, knights always told the truth and peasants sometimes lied and sometimes told the truth. The three spoke as follows:

A : B is a spy.
B : A is a knight.
C : Either A or B is telling the truth.
3. When a farm burned, a bill was found reading: 72 turkeys, \$.67.9_. The fire smudge obliterated the first and last digit of the total price of the birds. What are the two faded digits and what was the price of a single turkey?
Assumptions: Assume that the first digit cannot be 0. Assume each turkey costs the exact same and there's no deals. Assume the cost of a turkey requires the use of only whole pennies (so 1.023 isn't allowed).
Note: This problem is entirely solved by guessing and checking. There isn't anything really systematic going on here. Me choosing $y = 0$ was entirely arbitrary.

Stir it up This chapter is meant to teach you about different ways to attack problems when the "right" way isn't clear. The most familiar strategy will be that of **Trial and Error**. We will also consider **Experimentation**, what the book calls **Talk First, Then Think, Think Quantity, Work Backwards**, and **Analogous Problem**. We first consider the strategy of trial and error.

3 Trial and Error

Trial and Error

Guess and check—try different answers until you come up with the right one.

3.1 Cryptarithmic

To demonstrate the trial and error strategy, we will consider a type of cryptography problem.

Definition 1 (Cryptography). *Basically the study of breaking codes and encrypted messages.*

Definition 2 (Cryptarithmic). *An arithmetic puzzle where letters and arithmetical symbols are used to disguise an arithmetic computation.*

Examples 1. What does each letter in the following arithmetic equation mean: $ELF + ELF = FOOL$. We require that each letter represents a distinct digit from 0 to 9 and the leader letter to be nonzero (i.e., $F \neq 0$)

Solution 1. We start off by guess and check. Let's say $L=3$. Is it possible for $F+F$ to make $L=3$? In fact, can we ever make L an odd number? We see that L must be even because $L=2F$ or $1L=2F$. For example, if $F=4$, then $L=2F=8$ and if $F=6$, then $1L=2F=16$, and so $L=6$.

Let's guess for F and E . If $F=2$, do we have any issues? Can we ever choose E so that $E+E=2X$? No, so F must be 1.

So we now have $EL1 + EL1 = 1OOL$.

Then $1+1=L$ so we have $E21 + E21 = 1OO2$.

Then $2+2=O$ and so $E21 + E21 = 1442$.

Therefore $E=7$ and so $721 + 721 = 1442$

Okay, let's do another example.

Examples 2. $SHE + EEL = ELSE$

Solution 2. Again, we find immediately that $E=1$.

From this, $SH1 + 11L = 1LS1$.

Then $L=0$ so that $SH1 + 110 = 10S1$.

Then $S=9$ so that $9H1 + 110 = 1091$.

Lastly, $H=8$ so that $981 + 110 = 1091$.

Definition 3 (3×3 Magic Squares). *A 3 by 3 grid of squares such that when square is given a unique number from 1 to 9, the sums of the columns, rows, and two diagonals equal the same number.*

Examples 3. We will start off with an example. We first come up with several facts. Firstly, we note that since all the numbers between 1 and 9 will be in the squares, if we sum all the squares up, we get $1 + 2 + 3 + \dots + 9 = 9(10)/2 = 45$. Therefore, if we summed three columns, we get 45. Therefore, we must have that each column adds up to 15 since each column must add to the same number and 15 is the only number that when added three times gives us 45. Let's do some Stir it up. We place 9 in the center? No, because then we will always add over 15 for at least one row. Similar reasoning will show this is true for 6, 7, 8. Can we place 4 in the center? No, because we must place a 1 somewhere, giving us a sum greater than 15 somewhere. Similar reasoning will show this is true for 1, 2, 3. We conclude we must use 5 in the center. We conclude we must place 9 somewhere other than a corner. We solve the square!

Problem 1. *Find a magic square that is different than the one shown. How many magic squares are there?*

Examples 4. Consider a 4 by 4 magic square. We fill each square with a unique number from 1 to 16. Therefore, the sum of all the squares is $16(17)/2 = 136$. So since $136/4 = 34$, we conclude each row, column, diagonal, must add up to 34. So, in magic square theory, we consider symmetric magic squares to be equivalent; i.e., there is only one 3 by 3 magic square. For the 4 by 4 magic squares, there are 880. Here's a special one. Not only does every column, row, and diagonal add up to 34, so do the 2 by 2 squares inside.

3.2 Prisoner's Dilemma

Problem 2. *Two members of a criminal gang are arrested and imprisoned. Each prisoner is in solitary confinement with no means of communicating with the other. The prosecutors lack sufficient evidence to convict the pair on the principal charge. They hope to get both sentenced to a year in prison on a lesser charge. Simultaneously, the prosecutors offer each prisoner a bargain. Each prisoner is given the opportunity either to: betray the other by testifying that the other committed the crime, or to cooperate with the other by remaining silent. The offer is:*

- *If A and B each betray the other, each of them serves 2 years in prison*
- *If A betrays B but B remains silent, A will be set free and B will serve 3 years in prison (and vice versa)*
- *If A and B both remain silent, both of them will only serve 1 year in prison (on the lesser charge)*

Each prisoner is purely rational and self-interested. What is the best strategy for each prisoner? Assumption: They value certainty over chance.

Solution 3. Consider the case where A remains silent. Then we have two cases for B: either B remains silent or B betrays A. If B is silent, they both serve 1 year in prison. If B betrays A, then A serves 3 years and B goes free. We see that if A remains silent, he knows it is possible for him to serve 3 years of prison.

Now let's consider the case where A betrays B. Again, we have two cases for B: either B remains silent or B betrays A. If B is silent, then A gets set free and B serves 3 years—a better outcome than before. If B betrays A, then since both are betraying, they will both serve 2 years in prison.

We therefore see that if A chooses to betray, he knows for certain he will be better off than the worst possible case if he remained silent. Therefore, A is in fact better off choosing to betray B for he knows for certainty, the outcome isn't the worst.

Similarly, we can conclude B will choose to betray A.

Therefore, the best strategy for each prisoner is to betray the other.

3.3 More Trial and Error

Problem 3 (Ages). *Amy is 10 years older than Bill. In 5 years she will be twice as old. What are their two ages?*

Problem 4. *In 1996 my grandfather's age equaled the product of the four digits in the year he was born and the same was true for his daughter. Find the two birth years.*

4 Experiment

Another way to stir it up is by running an experiment and acquire data from the experiment to give you information and insight.

Problem 5 (Should You Switch?). *You are a contestant on a game show where one million dollars is hidden behind one of three closed doors. There is nothing behind either of the other two doors. You guess door 1, but before opening that door the game show host opens door 2 and shows you that nothing was hidden there. He offers you the chance to switch your guess to door 3. Should you switch?*

Solution 4. We will do an experiment as a class to gather some data. From this data, we will make a conjecture and then for your homework you will explain why the conjecture is true

The Experiment: We will run the following experiment as a class and record data along the way. For twenty trials we will do the following. I will choose a number from 1,2,3 and keep it secret. You will choose a number. I will tell you one of the numbers I haven't chosen. Then you will decide to not switch numbers. Then I will reveal the number and we will record the data.

Next, for twenty trials we will do the variation. Again, I choose a number from 1,2,3, and keep it secret. You choose a number. I will tell you one of the numbers I haven't chosen. Then you will decide to switch numbers. Then I will reveal the number and we will record the data.

From here we will be able gain an insight as to whether or not switching is beneficial.